Software Requirement Specification (SRS)
WEB BASED AUTOMATED MANUFACTURING SYSTEM (WAMS)

1. Introduction

1.1 Purpose

The Web Based Automated Manufacturing System (WAMS) is intended to help manufacturing companies manage product manufacturing based on dealer orders, maintain product and part inventory, handle supplier quotations, and perform billing and transaction management.
This document delineates the features of WAMS so as to serve as a guide to developers on one hand and a software validation document for management authorities and clients on the other.

1.2 Scope

We describe what features are in the scope of the software and what are not in the scope of the software to be developed.

<u>In Scope:</u>

a. Managing dealer information and dealer product orders.
b. Managing product stock and part stock.
c. Handling supplier details and quotations.
d. Processing supplier orders and manufacturing decisions.
e. Billing and product transaction management.
f. Generating reports for decision-making.
g. User authentication and role-based access.

<u>Out of Scope:</u>

a. Physical manufacturing operations on machines.
b. Logistics and transportation management.
c. Financial accounting and taxation systems.
d. Third-party ERP integrations.

1.3 Definitions, Acronyms, and Abbreviations

Acronyms and Abbreviations:

a. WAMS: Web Based Automated Manufacturing System
b. SRS: Software Requirements Specification
c. GUI: Graphical User Interface
d. DBMS: Database Management System

Definitions:

a. Dealer: An authorized entity that places product orders.
b. Supplier: An entity that supplies parts or raw materials.
c. Quotation: A price offer provided by a supplier.
d. Manufacturing Order: Authorization to produce products.
e. Product Transaction: Inflow or outflow of product stock.

1.4 References

IEEE Std 830-1998 / IEEE 29148-2018.
Database Management System lecture notes.
Software Engineering lecture notes.
Appendix A: Formula for Manufacturing Calculations.
Appendix B: Formula for Net Value Calculations.
Appendix C: Bill of Materials (BOM) Calculations.

1.5 Overview

The rest of this SRS is organized as follows:

Section 2 provides an overall description of the system, including user characteristics, constraints, and assumptions.
Section 3 specifies functional requirements using use cases.
Section 4 lists future extensions.
Section 5 contains appendices.

2. Overall Description

2.1 Product Perspective

WAMS is a web-based centralized system designed for manufacturing companies. It maintains a shared database accessible by different user roles such as administrators, managers, dealers, and suppliers

2.2 Product Functions

WAMS should support the following classes of use cases:

Class of use cases	Use cases	Description of use cases
Use case related to Installation	Installation	Creates and initializes working files.
Use cases related to system authorization	Login	Login to WAMS
	Signup	Signup to WAMS
	Logout	Logout from WAMS
	Change Password	Change WAMS Password
Use case related to User Management	Manage User Role	Assigns permissions to different system users (Admin, Staff, Manager).
	User Profile Management	Updates personal details and contact information of internal users.
Use case related to Dealer Management	Create Dealer Profile	New Dealer Registration
	Update or Delete Profile	Update or Delete existing dealer
	Dealer's order	Order placement
Use case related to Stock Management	Track Inventory	Provides real time visibility of raw parts and finished products.
	Update Stock	Adjusts the quantity of items after production or shipment.
	Set Reorder Levels	Triggers alerts when stock falls below a specific threshold.
Use case related to Supplier Management	Supplier detail	Store supplier Information
	Supplier Quotations	Manages inquiries sent to suppliers for raw material pricing.
Use case related to Manufacturing	Manufacturing Order generation	Generate and tracks manufacturing orders based on approved supplier order and stock requirement
	Manufacturing Completion & Stock Update	Involve the systematic process of transforming raw materials into finished goods and updating inventory records

		to reflect these changes in real time
Use case related to Billing and Payment	Bill Generating	Creates and initializes a formal billing document for a dealer.
	Payment Tracking	Monitors the payment status of outstanding invoices.
Use case related to Search and Filtering	Search/Filter Records	Allows users to quickly find specific orders, dealers, or stock items.
Use case related to Notifications	System Alerts	Sends automated notifications for low stock, new orders, or overdue payments.
Use case related to Reporting	Sales report, Stock report, Supplier report	Generate analytical report to support management decision-making and planning
Use case related to System Maintenance	Data Backup & Recovery	Ensures data safety by creating backups and allowing system restoration.

2.3 User Characteristics

a. **Administrator:** Technical staff managing system settings and user roles
b. **User:**Users should be familiar with basic manufacturing and inventory concepts.
c. **Manager:** Managers should understand sales and stock reports.
d. **Dealer:** Dealers should know product requirements and quantities.
e. **Supplier:** Supplier provides quotations and fulfills part orders.
f. **Management Authority**: Reviews reports and makes decisions

2.4 Principal Actors

The principal actors in WAMS are:
Administrator
User
Manager
Dealer
Supplier
System

2.5 General Constraints

a. Internet connection is required.
b. Role-based access must be enforced.
c. The system must use a relational database.

2.6 Assumptions and Dependencies

a. Suppliers respond to quotation requests on time.
b. Stock data entered is accurate.
c. Manufacturing decisions depend on reports generated by the system.

3. Specific Requirements

3.1 Functional Requirements

We describe the functional requirements by giving various use cases.

Use case related to installation:

Use Case 1: Installation
Primary Actor: User
Pre Condition: Internet connection available.

Main Scenario:

1. User initiates WAMS installation program.
2. **Configuration:**
   System prompts the user to select a **Home Directory** for data storage and to set up the **Primary Admin Credentials** (Login and Password).
3. **Path Selection:** User specifies the directory and enters the desired credentials.
4. **File Creation:** System creates the core database structure and working files in the specified directory.

   Working files contain:
   a. **Authorization & Role Matrix:** Initial security protocols and permission levels.
   b. **Inventory Master Ledger:** Files to track raw parts and finished goods.
   c. **Bill of Materials (BOM) Templates:** Structures for manufacturing recipes.
   d.**Supplier & Dealer Directories:** Initialized flat files for external entity records.
5. **Master Data Sync:** System connects to the central repository to download the **Global Raw Material Catalog** (Standard industry parts, SKU formats, and unit measurements). It initializes the system with default tax configurations and manufacturing unit standards.
6. **Success Confirmation:** System displays "Installation Complete" and prompts the user to log in for the first time.

Alternate Scenario :

3(a). Invalid Directory Path

3(a)1. System detects the directory is read-only or the path is invalid.
3(a)2. System displays an error: "Installation Path Not Accessible."
3(a)3. User is prompted to select a different location.
5(a). Network failure.
5(a)1. System detects a lost connection while downloading material catalogs.
5(a)2. System attempts to retry the connection three times.
5(a)3. If retries fail, the system prompts the user to "Finish without Sync" (Offline Mode) or "Abort Installation."
5(b). Database Initialization Error
5(b)1. A conflict occurs while creating the Authorization files.
5(b)2. System rolls back changes (deletes partial files) and displays a "Setup Failed" report.

Use cases related to system authorization :

Use Case 2: Signup
Primary Actor: User
Pre Condition: Application is installed.
Internet connection is available.

Main Scenario:

1. User launches the application.
2. User selects the **Sign Up** option..
3. System prompts for:
   3a. Username
   3b. Password
   3c. Confirm Password
   3d. Additional required details (if any)
4. User enters the requested information.
5. System validates the input data.
6. System creates a new user account.
7. System confirms successful registration and redirects the user to the main screen.

Alternate Scenario :

5(a). Username Already Exists
5(a)1. System displays an error message.
5(a)2. User is prompted to choose a different username.
5(b). Password Mismatch
5(b)1. System displays an error message.
5(b)2. User is allowed up to 3 attempts to re-enter passwords.
5(c). Invalid or Missing Data
5(c)1. System highlights invalid fields..
5(c)2. User is prompted to correct the information.

Use Case 3: Login
Primary Actor: User
Pre Condition: User is registered.

Main Scenario:

1. User starts the application.
2. System prompts for login ID and password.
3. User enters credentials.
4. System does authentication.
5. Main screen is displayed.

Alternate Scenario :

4(a). Authorization fails
4(a)1. Prompt the user that, he typed the wrong password/ID
4(a)2. Allow him to re-enter the password. Give him 3 chances.

Use Case 4: Logout
Primary Actor: User
Pre Condition: User logged in

Main Scenario:

1. User click on Logout button.
2. User prompted for logout
3. User gives the confirmation
4. System does Logout and terminates the session..
5. User redirected to login page.

Use Case 5: Change Password

Primary Actor: User
Pre Condition: User logged in

Main Scenario:

1. The user initiates the password change command.
2. User is prompted for:
   Old password
   New password
   Confirm new password.
3. User enters the required information.
4. The system does authentication and validates the old password.
5. System verifies password confirmation and a new password is registered.

Alternate Scenario :

4(a). Authorization fails/Incorrect Old Password

4(a)1. Prompt the user that he typed the wrong password
4(a)2. Allow him to re-enter the password. Give him 3 chances.
5(a). New password and confirm new password do not match.
5(a)1. User is allowed up to 3 attempts to re-enter new password details.

Use Case 6: Forgot Password

Primary Actor: User
Pre Condition: User is registered.

Main Scenario:

1. User selects **Forgot Password** on the login screen.
2. System prompts the user to enter registered username or email ID.
3. User enters the required information.
4. System verifies user details.
5. System sends a password reset link or temporary password to the registered email.
6. User resets the password.
7. System confirms password reset and redirects to the login screen.

Alternate Scenario :

4(a). User Not Found
4(a)1. System displays an error message.
4(a)2. User is prompted to re-enter valid details.
5(a). Email Delivery Failure
5(a)1. System displays an error message.
5(a)1. User is advised to retry later.

Use Cases Related to User management:

Use Case 7: Manage User Roles
Primary Actor: Admin
Pre Condition: Admin logged in into the system.

Main Scenario:

1. Admin navigates to the **User Management Console**.
2. System displays a list of all registered users, their status, and current roles.
3. Admin selects a specific user and chooses the "Edit Role" action.
4. System provides a dropdown of available roles:
   Admin
   Production Manager
   Inventory Manager
   Inventory Staff
   Sales Representative
   Auditor.

5. Admin selects the new role and submits.
6. System logs the change for auditing purposes and updates user permissions in real-time.

Alternate Scenario:

4(a). Self-Demotion Prevention:
Admin attempts to remove their own Admin rights while they are the only Admin.
4(a)1. "System prompt Error: Cannot demote the primary Administrator."
5(a). Database Conflict: Two Admins try to change the same user's role simultaneously.
5(a)1. System implements concurrency control and notifies the second Admin.
5(a)2. Record has been updated by another user. Please refresh.

Use Case 8: User Profile Management.
Primary Actor: User
Pre-Condition: User logged in.

Main Scenario:

1. User enters the "Settings" or "My Profile" section.
2. System displays current profile data: Name, Employee ID, Email, and Department.
3. User modifies the email address or phone number.
4. User clicks "Save Changes."
5. System validates the format of the new data and updates the record.

Alternate Scenario:

4(a). The profile whose name is supposed to change does not exist.
4(a)1. Renaming fails, the error message is displayed.
5(a). Profile with the same new user name exists.
5(a)1. Renaming fails, the error message is displayed.
5(b). Duplicate Email, same email is already used.
5(b)1. Email address is already in use, the error message is displayed.

Use Cases Related to Dealers

Use Case 9: Create Dealer Profile
Primary Actor: User
Pre Condition: User logged in.

Main Scenario:

1. User initiates the “create dealer profile” functionality.
2. System asks the user for the profile/dealer name.
3. User enters the profile name.

4. A dealer profile is created.

Alternate Scenario:

4(a). profile with the same name exists.
4(a)1. System asks the user for a different name.
4(a)2. User enters a different name.
4(a)3. New profile gets created.

Use Case 10:Update Dealer Profile.
Primary Actor: Dealer
Pre-Condition: Dealer is logged in.

Main Scenario:

1. User initiates the “rename portfolio” functionality.
2. System asks for the portfolio to be renamed and the new name.
3. User enters the new name.
4. Portfolio is renamed.

Alternate Scenario:

4(a). The portfolio whose name is supposed to change does not exist.
4(a)1. Renaming fails, the error message is displayed.
4(b). Portfolio with the same new name exists.
4(b)1. Renaming fails, the error message is displayed.

Use Case 11: Delete Dealer Profile.
Primary Actor: Dealer
Pre-Condition: Dealer is logged in.

Main Scenario:

1. User initiates the ”delete profile” functionality.
2. System asks for confirmation.
3. After confirming, Dealer profile will be deleted.

Alternate Scenario:

3(a). User cancel deletion process.
3(a)1. Deletion fails, error message is displayed.

Use Case 12: Dealer Order Management.
Primary Actor: Dealer
Pre-Condition: Dealer is logged in.

Main Scenario:

1. Dealer selects the "Place Order" module.
2. System displays a list of available products and their current lead times.
3. Dealer selects products, specifies quantities, and sets a preferred delivery date.
4. System validates the request and generates a "Dealer Request ID."
5. Manager receives an automated notification of the new order.

Alternate Scenario:

4(a). Dealer submits an order with missing quantities.
4(a)1. System highlights the empty fields and prompts for re-entry.
4(a)2. Dealer provides correct data, and the order is processed.

Use case related to Stock Management

Use Case 14: Track Inventory.
Primary Actor: User (Inventory Manager)
Pre-Condition: User logged in and has "Inventory Manager" permissions.

Main Scenario:

1. User navigates to the "Inventory Overview" dashboard.
2. System displays a real-time list of all raw parts and finished products.
3. System shows current quantity, location, and status (Available, Reserved, or In-Transit).
4. User applies a filter to view only "Raw Materials" or "Finished Goods."

Alternate Scenario:

2(a). Out of Sync Data:
User notices a discrepancy between physical count and system count.
2(a)1.User initiates a "Stock Audit" flag.
2(b)2. System logs the discrepancy for investigation.

Use Case 15: Update Inventory/Stock.
Primary Actor: User (Inventory Staff)
Pre-Condition: User logged in and is a staff.

Main Scenario:

1. User initiates the “Update Stock” functionality.
2. User selects the item and chooses the action:
   Add (Stock-In)
   Remove (Stock-Out)
   Adjust (Correction).
3. User enters the quantity and the reason

(e.g., "Shipment Received," "Damaged Goods," or "Manual Correction").
4. System updates the inventory levels and timestamps the change.

Alternate Scenario:

3(a). Negative Inventory Prevention:
User attempts to remove more stock than is currently available.
3(a)1. System blocks the transaction and displays:
"Error: Insufficient stock to fulfill this request."

Use Case 16: Set Reorder Levels.
Primary Actor: User (Inventory Manager)
Pre-Condition: User logged in and has "Inventory Manager" permissions.

Main Scenario:

1. User selects a specific raw material or finished product from the stock list.
2. User enters a "Minimum Threshold" value for that item.
3. System saves the reorder point.
4. When stock falls below this number, the system marks the item as "Low Stock."

Alternate Scenario:

2(a). Logical Error: User sets a threshold higher than what is currently in the warehouse
2(a)1.System saves the value but immediately triggers a "Low Stock" alert for that item.
2(b). User enters a negative number or non-numeric value.
2(b)1. System displays an error and prompts for a valid positive integer.

Use Cases Related to Supplier Management:

Use Case 18: Manage Supplier Details.
Primary Actor: Purchasing Officer/Admin
Pre-Condition: User logged in.

Main Scenario:

1. User initiates "Add/Update Supplier Profile."
2. User enters Supplier Name, Contact Info, Tax ID, and categories of materials supplied.
3. User sets "Lead Time" (the average time it takes for this supplier to deliver).
4. System saves the profile to the Supplier database.

Alternate Scenario:

4(a). Duplicate Supplier: System detects a supplier with the same Tax ID
4(a)1. System prompts:
"Supplier already exists. Would you like to update the existing record instead?"

Use Case 19: Manage Supplier Quotations.
Primary Actor: Purchasing Officer/Admin
Pre-Condition: Need for raw materials identified.

Main Scenario:

1. User (or System) initiates a "Request for Quotation" (RFQ).
2. System pulls a list of preferred suppliers for the specific raw material.
3. User selects the suppliers and sends the inquiry (Email/System Portal).
4. Suppliers submit their pricing and lead times.
5. User compares quotes in the **Supplier Management Dashboard**.
6. User selects a quote and clicks "Convert to Purchase Order (PO)."
7. System notifies the supplier and updates the "Incoming Stock" log.

Alternate Scenario:

3(a). No Suppliers Registered:

3(a)1. System alerts: "No suppliers found for this material. Please add a new supplier profile."

Use Cases Related to Manufacturing:

Use Case 20: Manufacturing Order (MO) Generation
Primary Actor: Production Manager
Pre-Condition: Approved Supplier Orders are present and Stock Requirements are calculated.

Main Scenario:

1. User initiates "Generate Manufacturing Order."
2. System prompts the user to select the **Finished Product** and the **Quantity** to be produced.
3. User enters the details.
4. System checks the **Bill of Materials (BOM)** to calculate the total raw materials needed for the requested quantity.
5. System compares needs against current stock.
6. If stock is sufficient, the system generates an MO with a unique ID and sets the status to "Ready for Production."
7. System generates a "Pick List" for the warehouse to move materials to the shop floor.

Alternate Scenario:

5(a). Shortage Conflict/Insufficient Raw Materials:
Required materials are currently "Reserved" for another order.

5(a)1. System alerts: "Material X is reserved for MO#Y.

Current production cannot proceed."

5(b). Stock Level Warning:

5(b)1. If the production of the finished product will exceed the warehouse storage capacity, the system issues a warning:
"Warning: Finished product storage exceeds set% of capacity."

6(a). Missing BOM Configuration:

6(a)1. If the selected product does not have a defined recipe/BOM, the system prompts:
"Error: No Bill of Materials found. Please configure product components before ordering."

6(b). Urgent Order Overwrite: Admin forces an MO to the top of the queue.

4(a)1. System re-calculates all other "Ready" orders to see if they now have a shortage due to the priority change.

Use Case 21: Manufacturing Completion (MC) & Stock Update.
Primary Actor: Production Manager/Shop Floor Supervisor
Pre-Condition: A Manufacturing Order (MO) is in "In-Progress" status.

Main Scenario:

1. User selects the active MO and marks it as "Completed."
2. User enters the final quantity produced (including any scrap/waste).
3. System performs a dual update:
   Deducts the actual raw materials used from the "Reserved Stock."
   Adds the finished products to the "Finished Goods Inventory."
4. System closes the MO and notifies the Sales/Dealer department that stock is ready for shipment.

Alternate Scenario:

2(a). Significant Production Variance:

2(a)1. If the finished quantity is significantly lower than the ordered quantity (due to defects), the system prompts for a "Reason Code" (e.g., Machine Failure, Material Defect).

3(a). Inventory Conflict:

3(a)1. If the system detects a mismatch between "Reserved" and "Physical" stock, it triggers an "Audit Alert" for the Inventory Manager.

Use Cases Related to Billing and Payment:

Use Case 22: Bill Generating
Primary Actor: Sales Manager / Accountant
Pre-Condition: Dealer order is marked as "Ready for Dispatch" or "Completed."

Main Scenario:

1. User initiates the “Generate Bill” functionality for a specific Dealer Order.
2. System retrieves order details: quantity, unit price, and dealer tax profile.
3. System calculates the total amount:

   Total = Sum (Quantity X Unit Price) + Taxes - Discount

4. System generates a unique **Invoice ID** and initializes a formal billing document (PDF/Digital format).
5. System updates the Dealer's account ledger with the "Outstanding Balance."

Alternate Scenario:

2(a). Missing Tax Configuration:
  2(a)1. System detects missing tax info for the dealer’s region.
  2(a)2. System prompts User to enter tax details before proceeding.
4(a). Partial Billing:
  4(a)1. User chooses to bill only for shipped items (Back-orders excluded).
  4(a)2. System generates a partial invoice and keeps the remaining order "Pending."

Use Case 23: Payment Tracking
Primary Actor: Sales Manager / Accountant
Pre-Condition: An active invoice exists in the system.

Main Scenario:

1. User navigates to the "Payment Tracking" module.
2. System displays a list of all outstanding invoices and their aging (e.g., 30, 60, 90 days).
3. User records a payment received (Cheque, Wire, or Cash).
4. System updates the invoice status to "Paid" or "Partially Paid."
5. System reconciles the dealer’s balance.

Alternate Scenario:

Use Cases Related to Notifications:

Use Case 25: System Alerts
Primary Actor: System (Automated)
Pre-Condition: A system trigger is met (Threshold hit).

Main Scenario:

1. **Low Stock Trigger:** System detects stock level $\le$ Reorder Level.
2. **New Order Trigger:** System detects a new dealer order entry.
3. **Overdue Trigger:** System detects an invoice past its due date.
4. System generates a notification badge and/or sends an email alert to the relevant department.

<u>Use Cases Related to Reporting:</u>

Use Case 26: Generate Analytical Reports
Primary Actor: Management / Admin
Pre-Condition: User logged in.

<u>Main Scenario:</u>

1. User selects the report type: **Sales Report, Stock Report, or Supplier Performance**.
2. User selects the time period (Daily, Monthly, Quarterly).
3. compiles data and generates a visual summary (Charts/Tables).
4. User exports the report as CSV or PDF.

<u>Alternate Scenario:</u>

3(a). Data Gap:

3(a)1. If no transactions occurred in the period, the system displays "No data available for the selected range."

<u>Use Cases Related to System Maintenance:</u>

Use Case 27: Data Backup & Recovery
Primary Actor: System Administrator / Admin
Pre-Condition: Admin logged in and System in "Maintenance Mode.".

<u>Main Scenario:</u>

1. Admin initiates "System Backup."
2. System creates a snapshot of the current database and working files.
3. System validates the backup file and stores it in the designated / Backup folder.
   (Recovery) Admin selects a backup file and clicks "Restore."
4. System overwrites current files with the backup data and restarts.

<u>Alternate Scenario:</u>

3(a). Backup Integrity Failure:

3(a)1. System detects corruption in the backup file.

3(a)2. System aborts the process and alerts: "Backup Failed: Check Disk Integrity."

3.2 Performance Requirements

(a) Hardware Compatibility: Should run on a machine with a 1.5 GHz processor and 4 GB RAM (standard for modern industrial workstations).
(b) Response Time: 95% of internal system responses (stock lookups, order status updates) should be within 1.5 seconds.
(c) Network Sync: For external supplier data syncing or high-volume report generation, a response time of up to 10 seconds is acceptable.

3.3 Design Constraints

1. **Data Security:** Access to sensitive manufacturing costs and dealer pricing must be restricted via Role-Based Access Control (RBAC). Database files must be encrypted to prevent unauthorized offline tampering.
2. **Fault Tolerance:** The system must implement "Atomic Transactions." If a power failure occurs during a stock update, the system must roll back to the last stable state to prevent "phantom inventory" (data corruption).
3. **Scalability:** The architecture should support a database growth of up to 100,000 SKUs without degradation in search performance.

3.4 External Interface Requirements

The User Interface (UI) is designed for high-efficiency warehouse operations. The screen is split into a **Navigation Sidebar** on the left and a **Work Canvas** on the right.

Left Pane: Contains the **Warehouse Tree**
(Organized by: Raw Materials => Work in Progress => Finished Goods).
Right Pane: Displays the detailed ledger, manufacturing order status, or dealer billing info for the item selected on the left.

4. Future Extensions

IoT Integration: Future versions will support RFID and Barcode scanner integration for automated stock updates.
Multi-Warehouse Support: Extending the software to manage multiple geographical warehouse locations from a single central server.

5. Appendix

5.1 Appendix A: Manufacturing Calculations

Formula for Total Manufacturing Cost (Cm): For a specific manufacturing order, the total cost is calculated by summing the raw materials, labor, and overhead:

$$Cm = \sum_{i=1}^{n} (Qi \times Pi) + (L \times R) + O$$

Where:

Qi = Quantity of raw material i
Pi = Purchase price of raw material i
L = Total labor hours
R = Hourly labor rate
O = Hourly labor rate

Formula for Reorder Point (RP):

$$RP\ (Average\ Daily\ Usage \times Lead\ Time) + Safety\ Stock$$

5.2 Appendix B: Net Value Calculations

Net Inventory Value (Raw Materials):

$$V_{raw} = \sum(StockOnHand_i \times UnitCost_i)$$

Net Worth of Outstanding Dealer Payments: The total liquid assets pending from dealers is calculated as:

$$W_{pending} = \sum(InvoiceAmount_j - AmountPaid_j)$$

5.3 Appendix C: Bill of Materials (BOM) Calculations

In a WAMS environment, the system must calculate the **Gross Requirement** for every component in a Bill of Materials when a Manufacturing Order (MO) is placed.

1. Basic Requirement Calculation
The total quantity of a specific raw material i required to produce a batch of finished products is calculated as:

$$R_i = Q_{order} \times q_i$$

Where:
R_i = Total required quantity of raw material i

Q_{order} = Total number of finished goods to be manufactured.
q_i = Quantity of material i required for a single unit (per the BOM recipe).

2. Accounting for Scrap/Wastage
In real-world manufacturing, materials are often lost during production (e.g., offcuts of steel, evaporated chemicals). To ensure enough material is issued, we apply a **Scrap Factor** (s):

$$Total\ Issue\ Quantity = (Q_{order} \times q_i) \div (1 - s_i)$$

Where:
s_i = The expected scrap rate for material i (expressed as a decimal, e.g., 0.05 for 5%).

3. Net Material Shortage (The Procurement Trigger)
To determine if we need to contact a supplier (linking to your **Supplier Management** module), the system calculates the **Net Shortage (S_{net})**:

$$S_{net} = R_i - (S_{avail} + S_{incoming})$$

Where:
S_{avail} = Current stock available in the warehouse.
$S_{incoming}$ = Stock already ordered from suppliers but not yet received.
Result: If $S_{net} > 0$, the system automatically triggers a **Supplier Quotation** request.